

Moral Hazard with Dual Risk-Averse Agent*

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Abstract

This paper studies a stylized moral hazard problem with a single agent protected by the limited liability constraint and is *dual risk-averse* in the sense of [Yaari \(1987\)](#). I consider the principal's cost-minimizing problem to induce a given action. Under some assumption on risk-aversion, which can be interpreted as the agent overweighting the probability of the worst outcome, and the strict version of the monotone likelihood ratio property, the solution to a relaxed problem where the agent's incentive compatibility constraint is replaced by its first-order condition is *debt contract*—the wage is constant up to some level of output, and then grows with the same speed as output afterwards. I also show that the first-order approach is valid if the outcome distribution is (in proportionate terms) less convex in action than the disutility of action, which is implied by [Rogerson \(1985\)](#)'s Convexity of Distribution Function Condition and convex disutility.

1 Introduction

Since the seminal work of [Holmström \(1979\)](#), an extensive literature on moral hazard has studied how a principal can incentivize an agent to take desirable actions when the agent's action is not contractible. A persistent critique of the theory is that it often predicts optimal contracts that are complex and highly sensitive to the underlying distribution of outcomes, in contrast to the simple contracts frequently observed in practice.

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This paper offers a new explanation for linear contracts by showing that *dual risk-aversion* implies the optimality of debt contracts. I consider a stylized moral hazard model with a single agent who is protected by the limited liability constraint and *dual risk-averse* in the sense of Yaari (1987); that is, he overweights the probabilities of “bad” outcomes when making decisions. I assume a specific form of dual risk-aversion where, if outcome follows distribution F , the agent behaves as if he believes that the worst outcome is realized with probability $(1 - k)$ and that outcome follows F with probability k , where $k \in (0, 1)$. I restrict the class of admissible contracts to those nondecreasing and with slope at most 1. To characterize the principal’s cost-minimizing contract for inducing each action, I first consider the *relaxed* problem, where the agent’s incentive compatibility constraint is replaced by its first-order condition, the so-called *FOA approach*. The relaxed problem is reduced to an optimal control problem, and Pontryagin’s maximum principle characterizes its solution. I show that, under the strict version of the monotone likelihood ratio property, the solution to the relaxed problem is *debt contract*; that is, wage is constant up to some level of outcome and grows at the same rate as outcome afterward (Proposition 1). I also show that this contract is indeed the solution to the original problem—the FOA approach is valid—if the outcome distribution is (in proportionate terms) less convex in action than the disutility of action (Proposition 2).

This paper contributes to the literature on moral hazard in two respects. First, I offer a new explanation, dual risk-aversion, for linear contracts observed in practice. Proposition 1 shows the optimality of debt contracts under the monotonicity and the limited liability constraints, which resonates with the finding of Innes (1990) and ?, who show the optimality of debt contracts under similar constraints.¹ Relatedly, Holmström and Milgrom (1987) and Carroll (2015) show the optimality of linear contracts in certain environments. Second, I derive a sufficient condition for validating the FOA. The seminal work of Rogerson (1985) shows that the FOA is valid if the distribution of outcome is convex in action (Convexity of Distribution Function Condition).² Recently, Chade and Swinkels (2020) establish a more general sufficient condition for validating the FOA. Leveraging their technique, I derive a sufficient condition in

¹? also show that a capped bonus contract, another form of linear contract, is optimal when the “critical ratio” is decreasing in the state.

²Since Convexity of Distribution Function Condition is restrictive, ? establishes an alternative condition and generalizes it to the case with more than one observable variable.

my environment (Proposition 2). As is the case in Chade and Swinkels (2020), this condition is implied by Convexity of Distribution Function Condition and convex disutility.

As an application of dual theory, this paper is also related to the set of papers, Gershkov, Moldovanu, Strack, and Zhang (2021, 2023); ?. Except Gershkov et al. (2021), they show that dual risk-aversion can better explain observed consumer behavior and forms of contracts than expected utility theory, which resonates with the message of this paper.

The remainder of the paper is organized as follows. Section 2 introduces the model. I characterize the solution to the relaxed problem in Section 3 and derive a sufficient condition for validating the FOA in Section 4. In Section 5, I discuss the implications of some modeling assumptions. Section 6 concludes the paper.

2 Model

Environment: I consider a stylized moral hazard environment with a single principal and a single agent.³ The agent takes a hidden action $a \in [0, \bar{a}]$ at a disutility $\psi(a)$, which is strictly increasing and C^2 . The disutility ψ is normalized so that $\psi(0) = 0$. To avoid corner solutions, I further assume the Inada conditions: $\psi'(0) = 0$ and $\psi'(\bar{a}) = +\infty$. The agent's action affects the distribution of outcome x , which takes a value in $[\underline{x}, \bar{x}]$. Given action a , the outcome x has a distribution $F(x | a)$ with everywhere positive density $f(x | a)$. I assume that F and f are C^2 and that the agent's action produces a first-order stochastic dominant shift on the outcome; that is, $F_a(x | a) < 0$ for all $x \in (\underline{x}, \bar{x})$ and all a .⁴

Preferences: The principal offers a bounded wage schedule $w: [\underline{x}, \bar{x}] \rightarrow [-m, M]$ to the agent, where $m, M \in \mathbb{R}_+$ and M is sufficiently large. She is risk-neutral and maximizes her expected net benefit $\mathbb{E}[x - w(x)]$. The agent is *dual risk-averse* in the sense of Yaari (1987): given the distribution of wage $\Gamma^w(\cdot | a)$ induced by wage schedule w and action a , his gross utility is given by

$$\int_{-m}^M \phi(1 - \Gamma^w(v | a)) dv, \quad (2.1)$$

³I use feminine pronouns for the principal and masculine pronouns for the agent.

⁴Note that $F_a(\underline{x} | a) = F_a(\bar{x} | a) = 0$ for all a since the support of the outcome is fixed.

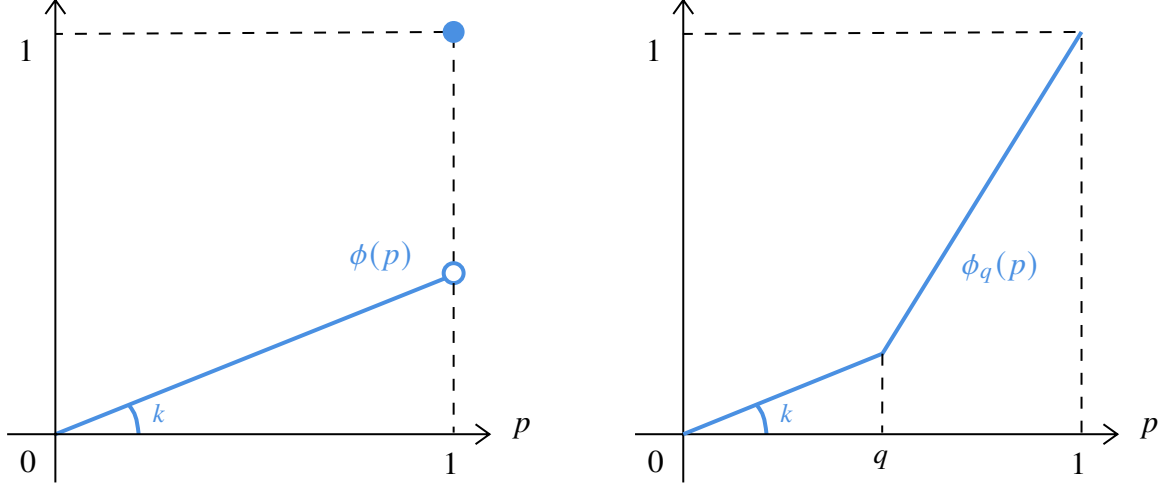


Figure 2.1: Distortion function ϕ can be interpreted as the limit of ϕ_q as q goes to 1.

where distortion function $\phi: [0, 1] \rightarrow [0, 1]$ is defined as

$$\phi(p) = \begin{cases} kp & \text{if } 0 \leq p < 1 \\ 1 & \text{if } p = 1 \end{cases}$$

for some $k \in (0, 1)$.

Remark 1. Yaari (1987) characterizes the dual representation for general continuous and nondecreasing ϕ satisfying $\phi(0) = 0$ and $\phi(1) = 1$, and risk-aversion is characterized by the convexity of ϕ . The present ϕ can be interpreted as the limit of the convex and piecewise-linear function that has a slope k on $[0, q)$ and extends from (q, kq) to $(1, 1)$. See Figure 2.1.

Admissible wage schedules: I assume $m = 0$ so that the agent is protected by the limited liability constraint. Following ?, I restrict attention to wage schedules that satisfy the following assumptions.

Assumption 1 (Monotonicity). $w(x)$ is nondecreasing.

Assumption 2 (Bounded Slope). $x - w(x)$ is nondecreasing.

As discussed in ?, Assumption 1 prevents the agent from sabotaging the task while Assumption 2 prevents the principal from doing so after the action has been chosen. The latter also prevents the agent from inflating the outcome by borrowing money to increase his returns. For $x \geq x'$, Assumptions 1 and 2 imply $w(x) \geq w(x')$ and $x - w(x) \geq x' - w(x')$,

respectively. By taking the limit, this implies that w is almost everywhere differentiable and Lipschitz continuous with

$$0 \leq w'(x) \leq 1 \quad \text{for almost every } x. \quad (\text{Lipschitz})$$

3 Principal's (relaxed) problem

Suppose that the principal wants to induce action $a > 0$. Given Assumption 1, the distribution of wage is given by $\Gamma^w(v | a) = F(w^{-1}(v) | a)$, where w^{-1} is the generalized inverse of w :

$$w^{-1}(v) = \sup\{x : w(x) \leq v\}.$$

The agent's gross utility (2.1) can be written as

$$\begin{aligned} \int_0^M \phi(1 - F(w^{-1}(v) | a)) dv &= w(\underline{x}) + \int_{\underline{x}}^{\bar{x}} \phi(1 - F(x | a)) w'(x) dx \\ &= w(\underline{x}) + \int_{\underline{x}}^{\bar{x}} k[1 - F(x | a)] w'(x) dx, \end{aligned}$$

where the first equality follows from the change of variable $x = w^{-1}(v)$. Integrating by parts, this can be further transformed into

$$(1 - k)w(\underline{x}) + k \int_{\underline{x}}^{\bar{x}} w(x) f(x | a) dx. \quad (3.1)$$

That is, the agent behaves as if he believes that the worst outcome $\underline{x}$ is realized with probability $1 - k$ and that outcome follows F with probability k . By doing so, the agent overweights the probabilities of bad outcomes in making his decision, which is consistent with Yaari (1987)'s interpretation of dual risk-aversion.

For the agent to accept wage schedule w and choose action a , his net utility $U(w, a)$ must be greater than or equal to his reservation utility U_0 :

$$U(w, a) \equiv w(\underline{x}) + \int_{\underline{x}}^{\bar{x}} k[1 - F(x | a)] w'(x) dx - \psi(a) \geq U_0. \quad (\text{PC})$$

Given w , he chooses action a if it maximizes $U(w, \tilde{a})$:

$$U(w, a) \geq U(w, \tilde{a}) \quad \text{for all } \tilde{a} \in [0, \bar{a}]. \quad (\text{IC})$$

Its first-order condition $U_a(w, a) = 0$ is equivalent to

$$\int_{\underline{x}}^{\bar{x}} k[-F_a(x | a)] w'(x) dx = \psi'(a). \quad (\text{FOA})$$

Since w is nondecreasing by **(Lipschitz)**, the limited liability constraint is reduced to

$$w(\underline{x}) \geq 0. \quad (\text{LL})$$

If the principal induces action a , she minimizes the expected payment $\int_{\underline{x}}^{\bar{x}} w(x) f(x | a) dx$ to the agent under the constraints **(PC)**, **(IC)**, **(LL)**, and **(Lipschitz)**. To characterize the solution to this problem, I first consider a relaxed problem where **(IC)** is replaced by **(FOA)**. Then, I derive a sufficient condition under which the solution to the relaxed problem indeed satisfies **(IC)**; that is, the FOA is valid. The relaxed problem can be formulated as follows:

$$\begin{aligned} \min \quad & \left[\int_{\underline{x}}^{\bar{x}} w(x) f(x | a) dx \right] \\ \text{subject to} \quad & w(\underline{x}) + \int_{\underline{x}}^{\bar{x}} k[1 - F(x | a)] w'(x) dx - \psi(a) \geq U_0 \quad (\text{PC}) \\ & \int_{\underline{x}}^{\bar{x}} k[-F_a(x | a)] w'(x) dx = \psi'(a) \quad (\text{FOA}) \\ & w(\underline{x}) \geq 0 \quad (\text{LL}) \\ & 0 \leq w'(x) \leq 1 \quad \text{for a.e. } x \quad (\text{Lipschitz}) \end{aligned}$$

Given **(Lipschitz)**, **(FOA)** can be satisfied for a only if

$$\int_{\underline{x}}^{\bar{x}} k[-F_a(x | a)] dx \geq \psi'(a). \quad (3.2)$$

In the remainder of the paper, I restrict attention to those actions satisfying (3.2) with a strict

inequality:⁵

$$A^* = \left\{ a \in (0, \bar{a}) : \int_{\underline{x}}^{\bar{x}} k[-F_a(x | a)] dx > \psi'(a) \right\}$$

To make the problem amenable to the control-theoretic technique, define α and β by

$$\begin{aligned} \alpha(x) &= w(\underline{x}) + \int_{\underline{x}}^x k[1 - F(\tilde{x} | a)] w'(\tilde{x}) d\tilde{x}, & \alpha'(x) &= k[1 - F(x | a)] w'(x), \\ \beta(x) &= \int_{\underline{x}}^x k[-F_a(\tilde{x} | a)] w'(\tilde{x}) d\tilde{x}, & \beta'(x) &= k[-F_a(x | a)] w'(x). \end{aligned}$$

The relaxed problem can be written as an optimal control problem (P) with states (w, α, β) and control w' :

$$\begin{aligned} \min \quad & \left[\int_{\underline{x}}^{\bar{x}} w(x) f(x | a) dx \right] \\ \text{subject to} \quad & w'(x) = u(x) \quad \text{for a.e. } x \\ & \alpha'(x) = k[1 - F(x | a)] u(x), \quad \beta'(x) = k[-F_a(x | a)] u(x) \quad (\text{P}) \\ & w(\underline{x}) \geq 0, \quad w(\bar{x}) \text{ free} \\ & \alpha(\underline{x}) = w(\underline{x}), \quad \alpha(\bar{x}) - \psi(a) \geq U_0, \quad \beta(\underline{x}) = 0, \quad \beta(\bar{x}) = \psi'(a) \\ & 0 \leq u(x) \leq 1 \quad \text{for all } x \in [\underline{x}, \bar{x}]. \end{aligned}$$

A wage schedule is called a *debt contract* if the wage is constant up to some level of outcome and grows at the same rate as outcome afterward.

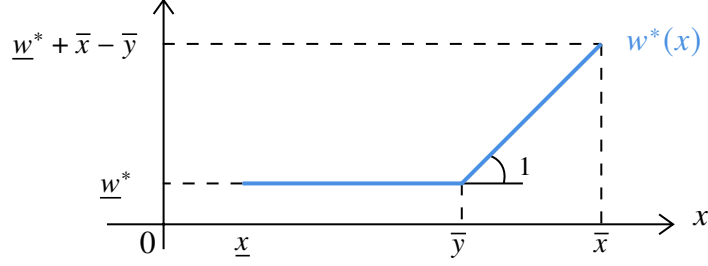
Definition 1. Wage schedule w is a *debt contract* if there exist y such that $w'(x) = 0$ if $x \in [\underline{x}, y)$ and $w'(x) = 1$ if $x \in (y, \bar{x}]$.

I show that, under the strict version of the monotone likelihood ratio property, the solution to (P) for each $a \in A^*$ is a debt contract.

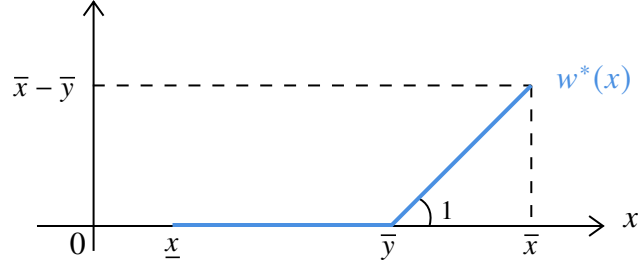
Assumption 3 (Strict MLRP). For each a , $\frac{\partial}{\partial x} \left(\frac{f_a(x|a)}{f(x|a)} \right) > 0$ for all x .

Proposition 1. Assume F satisfies Strict MLRP. For each $a \in A^*$, the cost-minimizing wage schedule w^* associated with the solution u^* to the relaxed problem (P) is debt contract.

⁵For action a that satisfies (3.2) with equality, the only way to induce a is to make the agent a residual claimant; that is, $w'(x) = 1$ for all x .



(i) When $\underline{w}^* > 0$



(ii) When $\underline{w}^* \leq 0$

Figure 3.1: The cost-minimizing wage schedule w^* is a debt contract.

Specifically, let $\underline{w}^* \equiv U_0 + \psi(a) - \int_{\bar{y}}^{\bar{x}} k[1 - F(x | a)] dx$, where $\bar{y}$ is the unique value satisfying $\int_{\bar{y}}^{\bar{x}} k[-F_a(x | a)] dx = \psi'(a)$. The cost-minimizing wage schedule w^* is given by

$$w^*(x) = \begin{cases} \max\{\underline{w}^*, 0\} & \text{if } x \in [\underline{x}, \bar{y}] \\ \max\{\underline{w}^*, 0\} + x - \bar{y} & \text{if } x \in (\bar{y}, \bar{x}]. \end{cases}$$

The solution w^* is depicted in Figure 3.1.

Given the implication of the distortion function ϕ I use, Proposition 1 is consistent with the literature. As I discussed at the beginning of this section, the agent behaves as if he is pessimistic, overweighting the probabilities of bad outcomes. Although he is risk-averse because of such pessimism, he can be considered “almost” risk-neutral because his net utility (3.1) is linear in wage. Indeed, Yaari (1987) discusses that risk-neutrality is characterized by ϕ being the identity function. Since the present ϕ is essentially an identity function scaled by k , it is reasonable to consider the agent almost risk-neutral. In this view, Proposition 1 resonates with Innes (1990), who shows the optimality of debt contracts under risk-neutrality, limited liability, and monotonicity constraints.

3.1 Proof of Proposition 1

In this section, I first define a class of simple contracts to simplify the exposition of the proof. I then characterize the solution to (P). The proof consists of two steps. In the first step, using Pontryagin's maximum principle, I solve (P) for fixed base wage $w(\underline{x})$. In the second step, I solve the optimization over $w(\underline{x})$.

3.1.1 Debt and capped bonus contract

Recall that in (P) the principal is restricted to wage schedules that satisfy (FOA):

$$\int_{\underline{x}}^{\bar{x}} k[-F_a(x | a)]w'(x) dx = \psi'(a). \quad (\text{FOA})$$

For each $a \in A^*$, by the implicit function theorem, there exists function $l_a : [\underline{x}, \bar{x}] \rightarrow [\underline{x}, \bar{x}]$ such that

$$\int_y^{l_a(y)} k[-F_a(x | a)] dx = \psi'(a).$$

I define *debt and capped bonus contract* w_{DB} as follows: for any $y \in [\underline{x}, \bar{y}]$, $\underline{w}$, and a , where $\bar{y} \equiv l^{-1}(\bar{x})$

$$w_{\text{DB}}(x | y, \underline{w}, a) = \begin{cases} \underline{w} & \text{if } x \in [\underline{x}, y] \\ \underline{w} + x - y & \text{if } x \in (y, l_a(y)] \\ \underline{w} + l_a(y) - y & \text{if } x \in [l_a(y), \bar{x}]. \end{cases}$$

That is, $w_{\text{DB}}(x | y, \underline{w}, a)$ is the wage schedule such that (i) the base wage is $\underline{w}$; (ii) the agent receives “bonus” on $[y, l_a(y)]$; and (iii) the wage is constant outside $[y, l_a(y)]$. In particular, for any $\underline{w}$ and a , $w_{\text{DB}}(x | \underline{x}, \underline{w}, a)$ is a *capped bonus contract*, and $w_{\text{DB}}(x | \bar{y}, \underline{w}, a)$ is a *debt contract* (see Figure 3.2).

For $w_{\text{DB}}(x | y, \underline{w}, a)$, (PC) can be written as

$$\underline{w} \geq U_0 + \psi(a) - \int_y^{l_a(y)} k[1 - F(x | a)] dx \equiv B(y).$$

Note that under Strict MLRP, the implicit function theorem also implies $B'(y) > 0$.

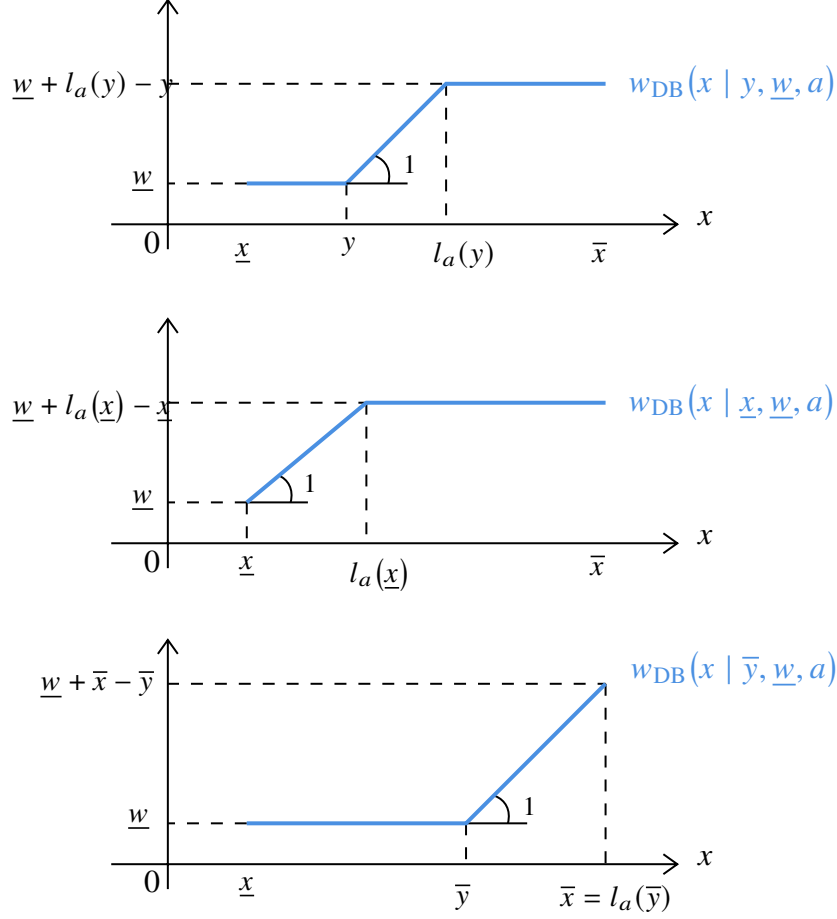


Figure 3.2: Debt and capped bonus contracts. Debt contract and capped bonus contract are special cases of this class.

3.1.2 Proof

First, I show that Strict MLRP implies that $\frac{f}{1-F}$ is decreasing in action, the property repeatedly used in the proof. Since $\frac{f_a}{f} = \frac{\partial}{\partial a} \ln f$, for any a and a' ,

$$\frac{f(x | a')}{f(x | a)} = \exp \left[\int_a^{a'} \frac{f_a(x | \tilde{a})}{f(x | \tilde{a})} d\tilde{a} \right].$$

Hence, by Strict MLRP, for any $x < x'$ and $a < a'$,

$$\frac{f(x' | a')}{f(x' | a)} > \frac{f(x | a')}{f(x | a)} \iff f(x' | a')f(x | a) > f(x | a')f(x' | a).$$

Integrating both sides with respect to x' from x to $\bar{x}$ and rearranging,

$$\frac{f(x | a)}{1 - F(x | a)} > \frac{f(x | a')}{1 - F(x | a')} \quad \text{for all } x \in [\underline{x}, \bar{x}]. \quad (3.3)$$

This proves that $\frac{f}{1-F}$ is decreasing in a .

Step 1: Solve (P) for fixed $w(\underline{x})$

Fix $w(\underline{x}) = \underline{w} \geq 0$. Define the Hamiltonian for (P) as follows:

$$H(w, u, p_1, p_2, p_3, x) = -p_0 w f + p_1 u + p_2 k(1 - F)u + p_3 k(-F_a)u.$$

Let $u^*(x)$ be a piecewise continuous control defined on $[\underline{x}, \bar{x}]$ that solves problem (P), and let $w^*(x)$, $\alpha^*(x)$ and $\beta^*(x)$ be the associated optimal paths.⁶ By Theorem 2 of Seierstad and Sydsaeter (1986, chap. 2), there exist constants p_0 , p_2^* , and p_3^* and a continuous and piecewise continuously differentiable adjoint function $p_1(x)$ such that, for all $x \in [\underline{x}, \bar{x}]$,

$$(p_0, p_1(x), p_2^*, p_3^*) \neq (0, 0, 0, 0) \quad (3.4)$$

$$p_0 = 0 \text{ or } p_0 = 1 \quad (3.5)$$

$$u^*(x) \text{ maximizes } H(w^*(x), u, p_1(x), p_2^*, p_3^*, x) \text{ for all } u \text{ in } [0, 1] \quad (3.6)$$

$$\text{Except at the points of discontinuities of } u^*(x), p_1'(x) = p_0 f(x | a) \quad (3.7)$$

$$p_1(\bar{x}) = 0, p_2^* \geq 0 \text{ and } p_2^*[\alpha^*(\bar{x}) - \psi(a) - U_0] = 0. \quad (3.8)$$

Since the Hamiltonian is linear and hence concave in (w, u) , the maximized Hamiltonian $\hat{H}(w, p_1, p_2, p_3, x) \equiv \max_{u \in [0, 1]} H(w, u, p_1, p_2, p_3, x)$ is also concave in w . Thus, by Theorem 5 of Seierstad and Sydsaeter (1986, chap. 2), the necessary conditions (3.4) to (3.8) are also sufficient for optimality if they are satisfied with $p_0 = 1$.

By (3.7) and (3.8), $p_1(x) = -p_0[1 - F(x | a)]$. Substituting this into the Hamiltonian,

$$H(w, u, p_1(x), p_2^*, p_3^*, x) = -p_0 w f - p_0(1 - F)u + p_2 k(1 - F)u + p_3 k(-F_a)u,$$

⁶I choose the value of $u(x)$ at a point of discontinuity x' as the left-hand limit of $u(x)$ at x' . I also assume that $u(x)$ is continuous at $\underline{x}$ and $\bar{x}$.

and hence

$$\begin{aligned} H_u^*(x) &\equiv \frac{\partial H}{\partial u}(w^*(x), u, p_1(x), p_2^*, p_3^*, x) \\ &= (p_2^*k - p_0)[1 - F(x | a)] + p_3^*k[-F_a(x | a)]. \end{aligned}$$

A necessary condition for (3.6) is

$$u^*(x) = \begin{cases} 1 & \text{if } H_u^*(x) > 0 \\ 0 & \text{if } H_u^*(x) < 0. \end{cases} \quad (3.9)$$

Suppose $p_0 = 0$. Then, $p_1(x) = 0$ for all x . Hence, by (3.4) and (3.8), at least one of $p_2^* > 0$ and $p_3^* \neq 0$ must hold.

Case 1-1: Suppose $p_2^* > 0$ and $p_3^* = 0$. Then, $H_u^*(x) = p_2^*k[1 - F(x | a)] > 0$ for all $x \in [\underline{x}, \bar{x}]$.

Hence, (3.9) implies $u^*(x) = 1$ for all x , which violates (FOA).

Case 1-2: Suppose $p_2^* = 0$ and $p_3^* \neq 0$. Since $H_u^*(x) = p_3^*k[-F_a(x | a)]$, either $H_u^*(x) > 0$ or $H_u^*(x) < 0$ for all $x \in (\underline{x}, \bar{x})$. Hence, (3.9) implies $u^*(x) = 1$ or $u^*(x) = 0$ for all x , which violates (FOA).

Case 1-3: Suppose $p_2^* > 0$ and $p_3^* \neq 0$. Since, $H_u^*(x) = p_2^*k[1 - F(x | a)] + p_3^*k[-F_a(x | a)]$, $H_u^*(x) > 0 \Leftrightarrow p_3^* > p_2^* \left[\frac{1-F(x|a)}{F_a(x|a)} \right] \equiv r(x)$. Note that (3.3) implies $r'(x) > 0$ and that $\lim_{x \rightarrow \underline{x}} r(x) = -\infty$. By (3.9), the unique candidate is a capped bonus contract:

$$w_{\text{DB}}(x | \underline{x}, \underline{w}, a) = \begin{cases} \underline{w} + x - \underline{x} & \text{if } x \in [\underline{x}, l_a(\underline{x})] \\ \underline{w} + l_a(\underline{x}) - \underline{x} & \text{if } x \in (l_a(\underline{x}), \bar{x}]. \end{cases}$$

By (3.8), (PC) must bind:

$$\underline{w} = B(\underline{x}).$$

Now suppose $p_0 = 1$. Then, $p_1(x) = -[1 - F(x | a)]$.

Case 2-1: Suppose $p_2^* = p_3^* = 0$. Then, $H_u^*(x) = -[1 - F(x | a)] < 0$ for all $x \in [\underline{x}, \bar{x}]$. Hence, (3.9) implies $u^*(x) = 1$ for all x , which violates (FOA).

Case 2-2: Suppose $p_2^* > 0$ and $p_3^* = 0$. Then, $H_u^*(x) = (p_2^*k - 1)[1 - F(x | a)]$. If $p_2^* \neq 1/k$, then (3.9) implies either $u^*(x) = 1$ or $u^*(x) = 0$ for all x , which violates (FOA). If $p_2^* = 1/k$, then any u^* is compatible with (3.9). In this case, (PC) must bind by (3.8).

In particular, the debt and capped bonus contract $w_{DB}(x | y, \underline{w}, a)$ satisfy all the necessary conditions if

$$\underline{w} = B(y).$$

Case 2-3: Suppose $p_2^* = 0$ and $p_3^* \neq 0$. Since $H_u^*(x) = -[1 - F(x | a)] + p_3^*k[-F_a(x | a)]$, $H_u^*(x) > 0 \Leftrightarrow p_3^* > \frac{1-F(x|a)}{k[-F_a(x|a)]} \equiv s(x)$. Note that (3.3) implies $s'(x) < 0$ and that $\lim x \rightarrow \underline{x} = +\infty$. By (3.9), the unique candidate is a debt contract:

$$w_{DB}(x | \bar{y}, \underline{w}, a) = \begin{cases} \underline{w} & \text{if } x \in [\underline{x}, \bar{y}] \\ \underline{w} + x - \bar{y} & \text{if } x \in (\bar{y}, \bar{x}], \end{cases}$$

with

$$\underline{w} \geq B(\bar{y}).$$

Case 2-4: Suppose $p_2^* > 0$ and $p_3^* \neq 0$. Since, $H_u^*(x) = (p_2^*k - 1)[1 - F(x | a)] + p_3^*k[-F_a(x | a)]$, $H_u^*(x) > 0 \Leftrightarrow p_3^* > \frac{(1-p_2^*k)[1-F(x|a)]}{k[-F_a(x|a)]} \equiv t(x)$.

- If $p_2^* = 1/k$, then $t(x) = 0$ for all x . Hence, (3.9) implies $u^*(x) = 1$ or $u^*(x) = 0$ for all x , which violates (FOA).
- If $p_2^* < 1/k$, then $\lim_{x \rightarrow \underline{x}} t(x) = +\infty$ and (3.3) implies $t'(x) < 0$. By (3.9), the unique candidate is a debt contract $w_{DB}(x | \bar{y}, \underline{w}, a)$ with binding (PC):

$$\underline{w} = B(\bar{y}).$$

- If $p_2^* > 1/k$, then $\lim_{x \rightarrow \underline{x}} t(x) = -\infty$ and (3.3) implies $t'(x) > 0$. By (3.9), the unique candidate is a capped bonus contract $w_{DB}(x | \underline{x}, \underline{w}, a)$ with binding (PC):

$$\underline{w} = B(\underline{x}).$$

Recall $B'(y) > 0$. Hence, for fixed $\underline{w}$, either (i) $\underline{w} = B(\underline{x})$; (ii) $\underline{w} = B(y)$ for some $y \in (\underline{x}, \bar{y})$; or (iii) $\underline{w} \geq B(\bar{y})$. Therefore, when $w(\underline{x})$ is fixed to $\underline{w}$, an optimal path w^* associated with a solution u^* is given by

$$w^*(x) = \begin{cases} w_{\text{DB}}(x \mid \underline{x}, \underline{w}, a) & \text{if } \underline{w} = B(\underline{x}) \\ w_{\text{DB}}(x \mid y, \underline{w}, a) & \text{if } \underline{w} = B(y) \\ w_{\text{DB}}(x \mid \bar{y}, \underline{w}, a) & \text{if } \underline{w} \geq B(\bar{y}). \end{cases} \quad (3.10)$$

Step 2: Optimization over $w(\underline{x})$

Given (3.10), I now solve the optimization over $\underline{w}$. Note that among the debt contracts $w_{\text{DB}}(x \mid \bar{y}, \underline{w}, a)$, the one with $\underline{w} = B(\bar{y})$ minimizes the expected payment to the agent. Hence, it is without loss to focus on $w_{\text{DB}}(x \mid y, B(y), a)$, and I consider the optimal choice of y for the principal. The principal's expected payment under this contract is

$$\begin{aligned} & \int_{\underline{x}}^y B(y) f(x \mid a) dx + \int_y^{l_a(y)} [B(y) + x - y] f(x \mid a) dx + \int_{l_a(y)}^{\bar{x}} [B(y) + l_a(y) - y] f(x \mid a) dx \\ &= U_0 + \psi(a) - \int_y^{l_a(y)} k[1 - F(x \mid a)] dx + \int_y^{l_a(y)} x f(x \mid a) dx \\ & \quad - y[1 - F(y \mid a)] + l_a(y)[1 - F(l_a(y) \mid a)] \\ &= U_0 + \psi(a) + (1 - k) \int_y^{l_a(y)} [1 - F(x \mid a)] dx, \end{aligned}$$

where the second inequality follows from applying integration by parts to the fourth term. The implicit function theorem implies that the third term in the last line is decreasing in y . Thus, the principal always chooses a debt contract ($y = \bar{y}$). Letting $\underline{w}^* = B(\bar{y})$, the cost-minimizing wage schedule w^* is given by

$$w^*(x) = w_{\text{DB}}(x \mid \bar{y}, \max\{\underline{w}^*, 0\}, a) = \begin{cases} \max\{\underline{w}^*, 0\} & \text{if } x \in [\underline{x}, \bar{y}] \\ \max\{\underline{w}^*, 0\} + x - \bar{y} & \text{if } x \in (\bar{y}, \bar{x}]. \end{cases}$$

4 The validity of the first-order approach

I now turn to the validity of the FOA. Leveraging the technique in [Chade and Swinkels \(2020\)](#), I derive a sufficient condition under which the agent's net utility is strictly quasiconcave in action, which implies that the condition (FOA) characterizes the optimal choice of action. As in [Chade and Swinkels \(2020\)](#), the concavities (in proportionate terms) of outcome distribution F and disutility ψ are the keys to validating the FOA.

Proposition 2. *Under Strict MLRP, the first-order approach is valid if*

$$\frac{F_{aa}(x | a)}{F_a(x | a)} < \frac{\psi''(a)}{\psi'(a)} \quad \text{for all } x \in (\bar{y}, \bar{x}). \quad (4.1)$$

Proof. The proof follows almost the same steps as [Chade and Swinkels \(2020, Proposition 5\)](#).

Recall that the agent's net utility from wage schedule w and action $\tilde{a}$ is given by

$$U(w, \tilde{a}) = w(\underline{x}) + \int_{\underline{x}}^{\bar{x}} k[1 - F(x | \tilde{a})] w'(x) dx - \psi(\tilde{a}).$$

The first- and second-order derivatives with respect to action are

$$\begin{aligned} U_a(w, \tilde{a}) &= \int_{\underline{x}}^{\bar{x}} k[-F_a(x | a)] w'(x) dx - \psi'(a) \\ U_{aa}(w, \tilde{a}) &= \int_{\underline{x}}^{\bar{x}} k[-F_{aa}(x | a)] w'(x) dx - \psi''(a), \end{aligned}$$

If $U_a(w, a) = 0$,

$$\psi''(a) = \frac{\psi''(a)}{\psi'(a)} \psi'(a) = \int_{\underline{x}}^{\bar{x}} \frac{\psi''(a)}{\psi'(a)} k[-F_a(x | a)] w'(x) dx.$$

Using this equation, $U_{aa}(w, a)$ can be written as

$$U_{aa}(w, a) = -k \int_{\underline{x}}^{\bar{x}} \left[F_{aa}(x | a) - \frac{\psi''(a)}{\psi'(a)} F_a(x | a) \right] w'(x) dx.$$

Note that $U_{aa}(w, a) < 0$ suffices for $U(w, \cdot)$ to be strictly quasiconcave. For w^* , this condition is

reduced to

$$\int_{\bar{y}}^{\bar{x}} \left[F_{aa}(x | a) - \frac{\psi''(a)}{\psi'(a)} F_a(x | a) \right] dx > 0,$$

which holds if

$$\frac{F_{aa}(x | a)}{F_a(x | a)} < \frac{\psi''(a)}{\psi'(a)} \quad \text{for all } x \in (\bar{y}, \bar{x}).$$

□

The condition (4.1) requires that for all $x \in (\bar{y}, \bar{x})$, the outcome distribution F is (in proportionate terms) less convex in action than the disutility ψ . In particular, (4.1) is satisfied if both F and ψ are convex in action, as in Rogerson (1985).

5 Discussion

In this section, I discuss some alternative modeling assumptions.

5.1 Distortion function ϕ

While Yaari (1987) characterizes dual risk-aversion by the convexity of ϕ , I have so far assumed a specific convex and *piecewise-linear* ϕ that has a natural interpretation. Another simplest possible ϕ is a piecewise-linear function that stays at 0 until q and extends from $(q, 0)$ to $(1, 1)$:

$$\phi(p) = \begin{cases} 0 & \text{if } 0 \leq p \leq q \\ \frac{p-q}{1-q} & \text{if } q < p \leq 1 \end{cases}$$

for some $q \in (0, 1)$. See Figure 5.1. This ϕ also has a natural interpretation. It is easy to see that the agent's gross utility can be written as

$$w(\underline{x}) + \int_{\underline{x}}^{x(a)} \frac{1 - F(x | a) - q}{1 - q} w'(x) dx, \quad (5.1)$$

where $x(a) \equiv F^{-1}(1 - q | a)$. That is, the agent behaves as if he is so pessimistic that he ignores the possibilities of good outcomes $x > x(a)$. Note that a debt contract can never be optimal in this case; since the agent virtually ignores outcomes $[x(a), \bar{x}]$, it is optimal for the principal to

set a constant wage in this interval. Indeed, by following similar steps as in Section 3, I can show that the cost-minimizing wage schedule features a constant wage for high (and low) outcomes, and that the wage grows at the same rate with outcome elsewhere.

This observation can be further generalized to a general convex ϕ . Specifically, I can show that the cost minimizing wage schedule must have bang-bang property; that is, the slope $w'(x)$ is either 0 or 1 for almost every x . Bang-bang property results from the fact that the Hamiltonian is linear in control, whatever ϕ I use, and hence is a robust property of the solution.

5.2 Bounded slope

Following ?, I have assumed Monotonicity and Bounded Slope of wage schedules (Assumptions 1 and 2). On one hand, they have some rationales such as deterring sabotage by either the principal or the agent. On the other hand, they jointly imply Lipschitz continuity of wage schedules, thereby excluding any discontinuity. In particular, wage schedules with upward jumps are not admissible.

The principal’s problem is well defined even without the Bounded Slope assumption. A simple calculation can show that a two-part wage schedule, which pays a flat base-wage until the threshold and pays a flat bonus-wage afterward, can induce the same action at a smaller cost than the cost-minimizing schedule characterized in Section 3. Now that upward jumps are admissible, by letting the wage jump up at a certain threshold, the principal can give the agent enough incentive without creating a “bonus region” as in the case of a debt contract. This helps reduce the expected payment to the agent, thereby achieving a smaller agency cost.

6 Conclusion

This paper has explored the implications of dual risk-aversion in a standard moral hazard setting with monotonicity and limited liability constraints. Under a specific form of risk-aversion where the agent overweights the probability of bad outcomes, the cost-minimizing wage schedule takes a particularly simple form, a debt contract. Given that the agent is considered almost risk-neutral, this result resonates with Innes (1990), who shows the optimality of debt contracts under similar

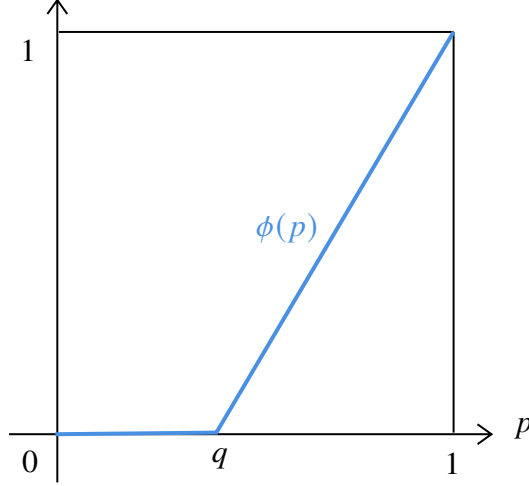


Figure 5.1: Another piecewise-linear distortion ϕ .

constraints. Leveraging the technique developed in [Chade and Swinkels \(2020\)](#), I have also derived a sufficient condition for validating the FOA. As is consistent with [Chade and Swinkels \(2020\)](#), the condition involves the concavities (in proportionate terms) of outcome distribution and disutility of action and is implied by [Rogerson \(1985\)](#)'s Convexity of Distribution Function Condition and convex disutility.

There are some extensions worth exploring. First, it is of theoretical interest to fully characterize the shape of the cost-minimizing wage schedule under a general distortion function. Although the bang-bang property can be easily shown, it is not yet clear how dual risk-aversion interacts with the principal's incentive and shapes the cost-minimizing wage schedule. Second, I have assumed the monotonicity of wage schedules throughout the paper. It has some rationale, such as preventing the agent from sabotaging the task. However, the monotonicity of the solution is indeed one of the results in the seminal work of [Holmström \(1979\)](#) and the follow-up papers. It is thus worthwhile to explore whether the monotonicity can be obtained through the principal's optimization problem.

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